

7. Mathematical and Generalized Models

One-line summary

This chapter organizes the observed relationships among candidate count, candidate reduction, performance-source contamination, Auto Arrange, and final order through provisional mathematical models. The models describe an observation space; they do not prove the game's internal algorithm.

Boundary statement. Candidate generation, subset selection, and final ordering are convenient analytical stages. They are not claims that the game implements three corresponding internal procedures. Random-number generation, weights, priority rules, exact processing order, and reselection remain unresolved.

7-1. Generalized candidate-count model

Let:

- N = total number of candidates that may compete for arrangement slots;
- M = maximum number of adopted arrangement slots (normally three);
- W = number of candidate subsets that satisfy the stated success condition;
- p = single-attempt success probability under the stated condition;
- $E = 1/p$ = expected number of attempts until one success.

Candidates may originate from inherited equipment, directly added materials, equipment-internal arrangements, Auto Arrange, performance-source contamination, or nested references. These labels identify observed sources; they do not establish an internal candidate-generation algorithm.

For the ordinary one-step recipes examined here, N is often no greater than about seven. This is a practical observation range, not a verified internal maximum.

The number of possible adopted subsets is:

$$K(N,M) = C(N, \min(N,M))$$

Therefore:

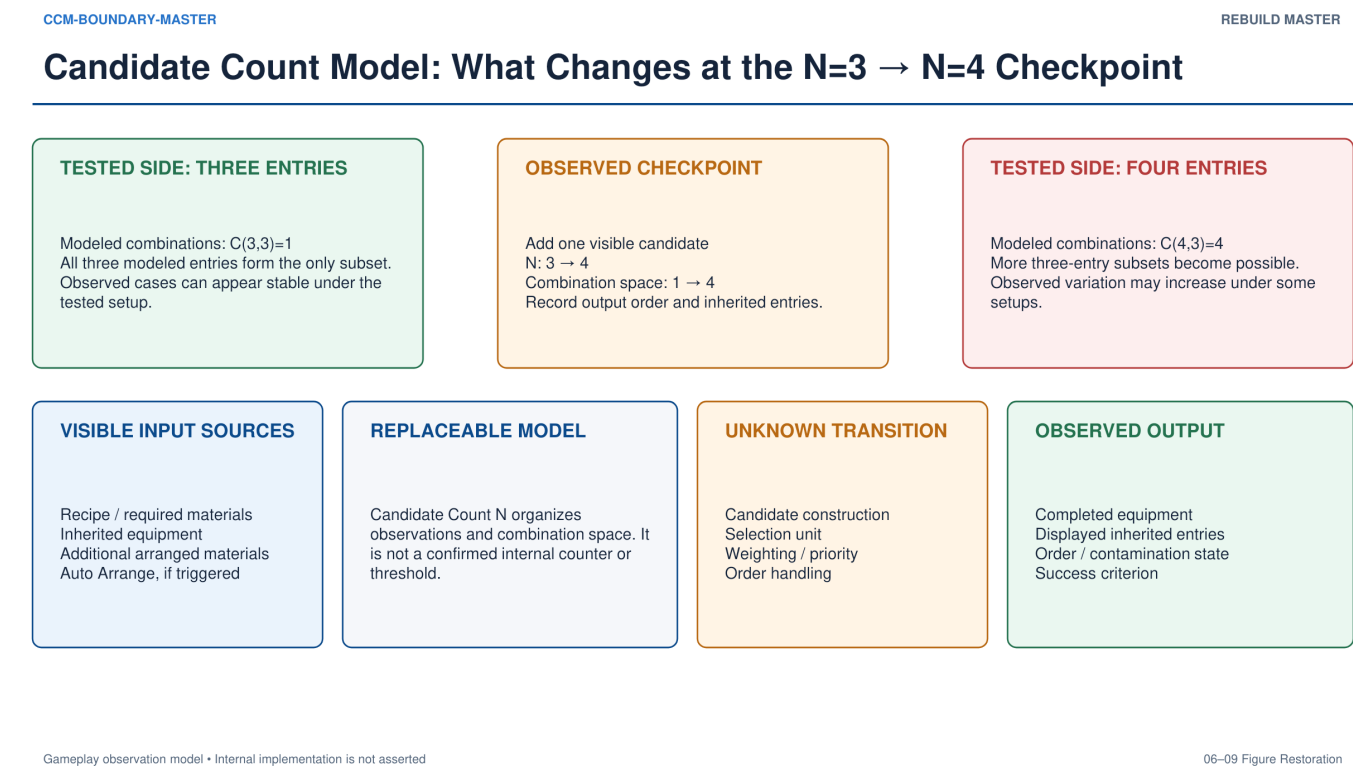
- if $N \leq M$, all candidates can fit and $K(N,M) = C(N,N) = 1$;
- if $N > M$, candidate reduction is required and $K(N,M) = C(N,M)$.

For $M = 3$, four distinct candidates produce four possible subsets: ABC, ABD, ACD, and BCD.

This equation determines the **size of the combination space**. It does not determine how probability is distributed over that space.

Figure 07-P04-img-000

Master: CCM-BOUNDARY-MASTER | Source anchor: Japanese Canonical p.4

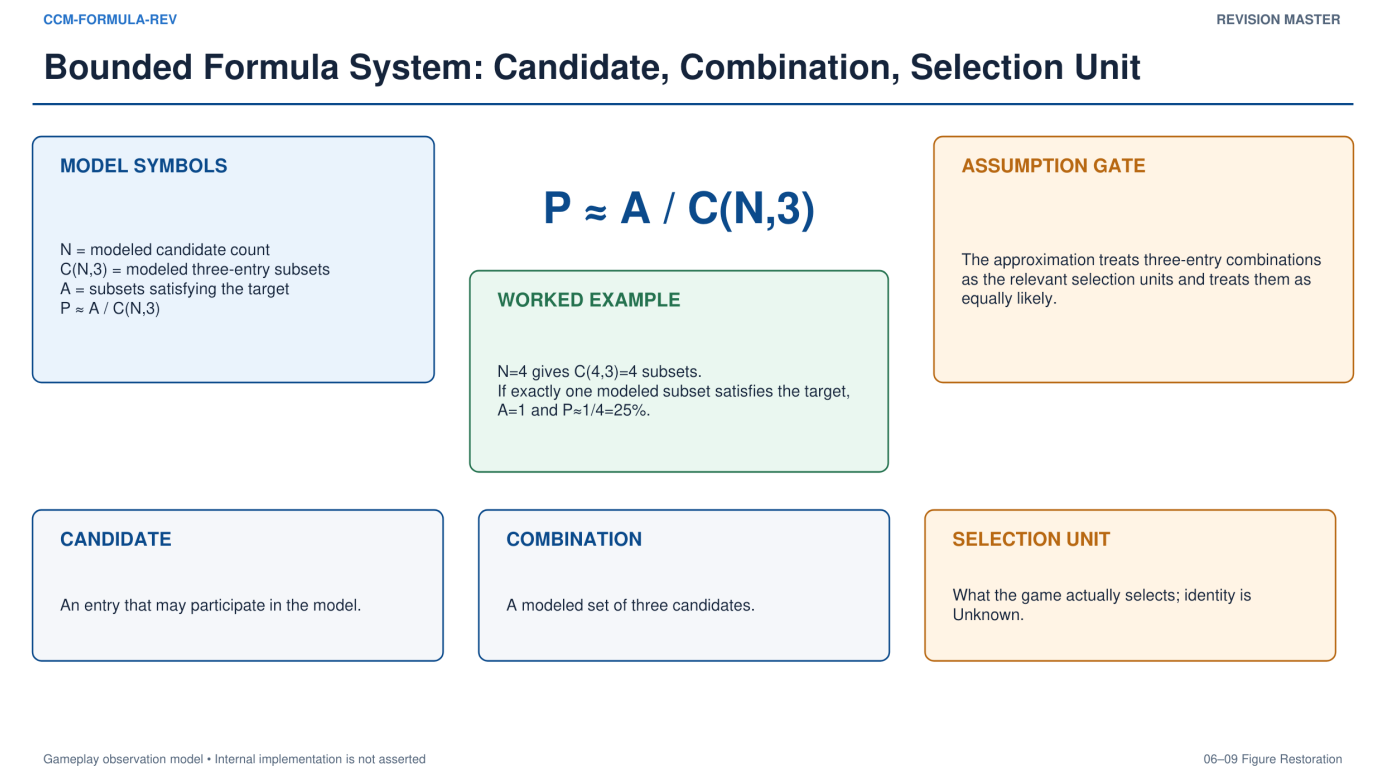


Placement caption

N<=3 / N>=4 boundary concept
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 07-P05-img-001

Master: CCM-FORMULA-REV | Source anchor: Japanese Canonical p.5



7-2. Conditional success-probability approximation

If, and only if, all candidate subsets are treated as equally likely and no duplicate-handling complication applies, success may be approximated by:

$$P(\text{subset success}) \approx W / K(N,M)$$

For one winning three-candidate subset and $M = 3$:

Candidate count N	Possible subsets C(N,3)	Equal-selection approximation
3 or fewer	1	100% candidate retention
4	4	25%
5	10	10%
6	20	5%
7	35	2.86%

The Japanese Canonical table listed 20%, 15%, and 11.43% for $N = 5, 6, 7$. Those values conflict with both $1/C(N,3)$ and the Canonical's own intermediate calculations. The English edition corrects them to 10%, 5%, and approximately 2.86%.

Observation update: equal selection is not guaranteed

The RF4SP turnip-family test produced candidate-subset frequencies 0 / 5 / 42 / 3 over 50 trials (chi-square = 93.84, $df = 3$, Cramer's $V \approx 0.791$). This rejects the auxiliary assumption that the four observed subsets are selected at 25% each under that condition.

It does **not** falsify the Candidate Count Model as a model of combination-space growth. The cause of the skew—material-specific behavior, registration order, candidate-generation order, or another internal process—remains **Unknown**.

By contrast, the RF4SP Messhilite 400-trial dataset has not shown the same type of subset skew in the current analysis. Its detailed conditions and results belong to Chapter 8.

7-3. Expected attempts and practical success thresholds

For a single-attempt success probability p :

$$E = 1 / p$$

After k independent attempts, the probability of at least one success is:

$$P(\text{at least one success}) = 1 - (1-p)^k$$

The expected-attempt count is not a high-confidence completion point. For example, at $p = 1/4$, $E = 4$, but four attempts produce only $1-(3/4)^4 \approx 68.4\%$ probability of at least one success.

Master: COMBINATION-COUNT-CURRENT | Source anchor: Japanese Canonical p.8

COMBINATION-COUNT-CURRENT

Rune Factory 5

Candidate Count → Success Probability Model

Why more candidates make success harder

Candidate Sources	Combination Explosion	Key Rule
Different sources add more candidates to the pool - Required Materials - Inherited Equipment - Auto Arrange - Self Contamination - Recursive Processing - Meshlike Expansion ↓ Candidate Pool (N) Total number of available candidates.	Combination Explosion The combination space grows exponentially as candidate increases! N = 4 Possible 3-Item combinations 4 combinations N = 8 Possible 3-Item combinations 56 combinations Valid Combinations (A) Only a small portion are valid ↓ Success Probability $P = \frac{A}{C(N,3)}$ A: Number of valid successful combinations	Key Rule Candidate Count ↑ ↓ Combination Count grows rapidly ↓ If A (valid combinations) stays similar Success Rate Falls ↓ Expected Attempts Increase ↓ Player Cost Increases
⇒ Key Insight ⇐ Increasing candidate count alone does not improve success. Success increases only when the number of valid combinations grows faster than the total combination space.		

The source figure retains all four subsets — ABC, ABD, ACD, BCD — and the worked count $C(4,3)=4$.

The 25% result is an equal-combination model example, not proof of the game's selection unit or weighting.

SOURCE DETAIL + CURRENT BOUNDARY

Gameplay observation model • Internal implementation is not asserted

06-09 Figure Restoration

Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Current English integration - figure authority migrated from the Japanese Canonical inventory.

When $E = 1/p$ is used as the scale:

Attempts	Approximate chance of at least one success
E	about 63% for small p
3E	about 95%
4.6E	about 99%

The Japanese column used N for both candidate count and reciprocal success probability. The English edition separates them: candidate count is N , probability is p , and the attempt scale is $E = 1/p$. Candidate count alone does not determine E unless a specific probability model has also been justified.

If one attempt takes t minutes, the rough work-time estimate is attempts $\times t$. Split saves do not change p ; they reduce the amount of work lost after failure. Three consecutive 25% stages have a through-success probability of $1/64$, whereas saving after each successful stage changes the expected workload from 64 complete sequences to roughly $4 + 4 + 4 = 12$ stage attempts.

7-4. Boundary around the three-slot limit

In the observed RF5 conditions, $N \leq 3$ often behaved as a retention region because all candidates could fit within the three arrangement slots. At $N \geq 4$, candidate reduction became necessary and instability was more frequently observed.

This boundary is associated with the observed three-slot limit. It is not proof of a hard-coded internal threshold. Occupied internal arrangement slots, Auto Arrange, performance-source contamination, and nested references may change the effective competition condition.

The key distinction is:

- $N \leq M$: no candidate needs to be dropped merely because of slot capacity;
- $N > M$: at least one candidate cannot be retained;
- neither statement determines final order;
- neither statement proves equal selection.

7-5. Equal-selection assumption

Equal selection is a useful baseline, not a confirmed mechanism. It assumes that every eligible subset receives the same probability. Existing-arrangement retention, Auto Arrange displacement, candidate-specific skew, or conditional processing could all produce a different distribution.

Accordingly:

Master: CUMULATIVE-PROBABILITY-CURRENT | Source anchor: Japanese Canonical p.15

REUSE / TRANSLATE

Success Probability Curve

As the candidate pool expands, successful combinations become increasingly diluted, reducing the probability of obtaining a specific inheritance result.

The graph illustrates the relationship between the number of candidates (N) and the success probability. The y-axis represents Success Probability on a logarithmic scale from 0.1% to 100%. The x-axis represents the Candidate Count (N) from 3 to 8. The curve is divided into three regions: Stable Region (N=3 to 4), Candidate Explosion (N=4 to 6), and Combination Dilution (N=6 to 8). The probability drops significantly as N increases, especially in the Candidate Explosion and Combination Dilution regions.

Candidate Count (N)	Combinations	Success Probability (C(N,3))
3	1 Combination	100%
4	4 Combinations	25%
5	10 Combinations	10%
6	20 Combinations	5%
7	35 Combinations	2.5%
8	56 Combinations	1.25%

More Candidates
The candidate pool becomes larger.

More Possible Combinations
The number of possible combinations increases exponentially.

Lower Chance
The chance of obtaining a specific combination drops rapidly.

Key Takeaway

Increasing the candidate count expands the combination space much faster than most players intuitively expect. As a result, even when the desired inheritance remains possible, its relative probability rapidly decreases because it occupies a smaller fraction of all possible combinations.

Note: $N = 8$ represents a highly specialized inheritance scenario that generally requires intentional recursive inheritance planning and is primarily applicable to accessories and footwear. It is included here only as a conceptual example illustrating how rapidly the combination space expands as candidate count increases.

Conceptual Illustration based on the Candidate Count Model.
The curve illustrates the general relationship between candidate count and inheritance success probability and is not intended to represent exact gameplay values.

The source figure retains the probability curve, trial multipliers, expected-attempt context, and planning interpretation.

Mathematical planning chart only. Fixed p and independence are explicit assumptions; inheritance mechanism is outside this figure.

SOURCE DETAIL + CURRENT BOUNDARY

06-09 Figure Restoration

Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Current English integration - figure authority migrated from the Japanese Canonical inventory.

Master: CUMULATIVE-PROBABILITY-CURRENT | Source anchor: Japanese Canonical p.16

REUSE / TRANSLATE

Success Probability Curve

As the candidate pool expands, successful combinations become increasingly diluted, reducing the probability of obtaining a specific inheritance result.

The graph illustrates the relationship between the number of candidates (N) and the success probability of obtaining a specific inheritance result. The Y-axis represents Success Probability (log scale, 0.1% to 100%), and the X-axis represents Candidate Count (N, 3 to 8). The curve shows a sharp decline in probability as N increases.

Candidate Count (N)	Number of Combinations	Combination Formula
3	1	$C(3,3) = 1$
4	4	$C(4,3) = 4$
5	10	$C(5,3) = 10$
6	20	$C(6,3) = 20$
7	35	$C(7,3) = 35$
8	56	$C(8,3) = 56$

The graph is divided into three regions:

- Stable Region:** Small increase in N has limited impact.
- Candidate Explosion:** Combination count grows exponentially.
- Combination Dilution:** Desired combinations occupy a tiny fraction of the whole.

Below the graph, three icons illustrate the concepts:

- More Candidates:** The candidate pool becomes larger.
- More Possible Combinations:** The number of possible combinations increases exponentially.
- Lower Chance:** The chance of obtaining a specific combination drops rapidly.

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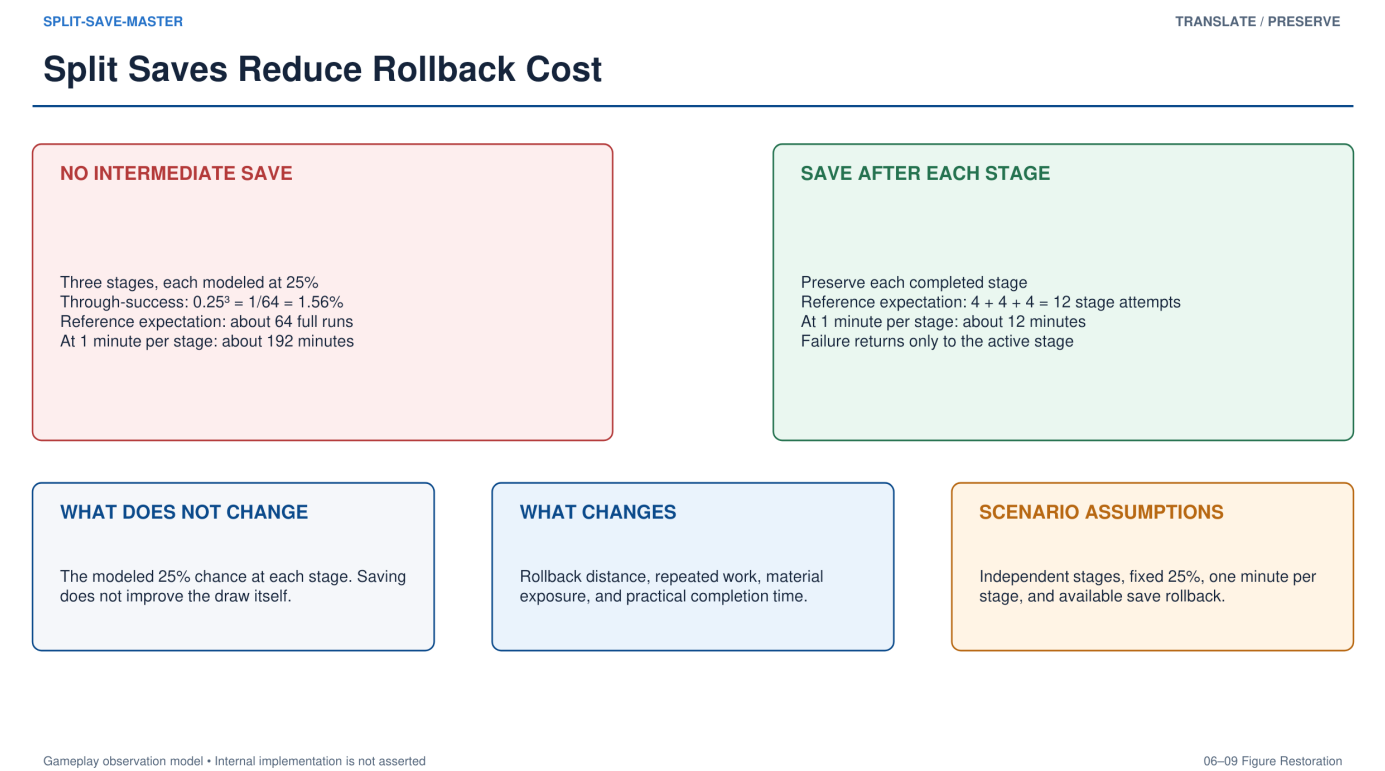
Mathematical planning chart only. Fixed p and independence are explicit assumptions; inheritance mechanism is outside this figure.

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Current English integration - figure authority migrated from the Japanese Canonical inventory.

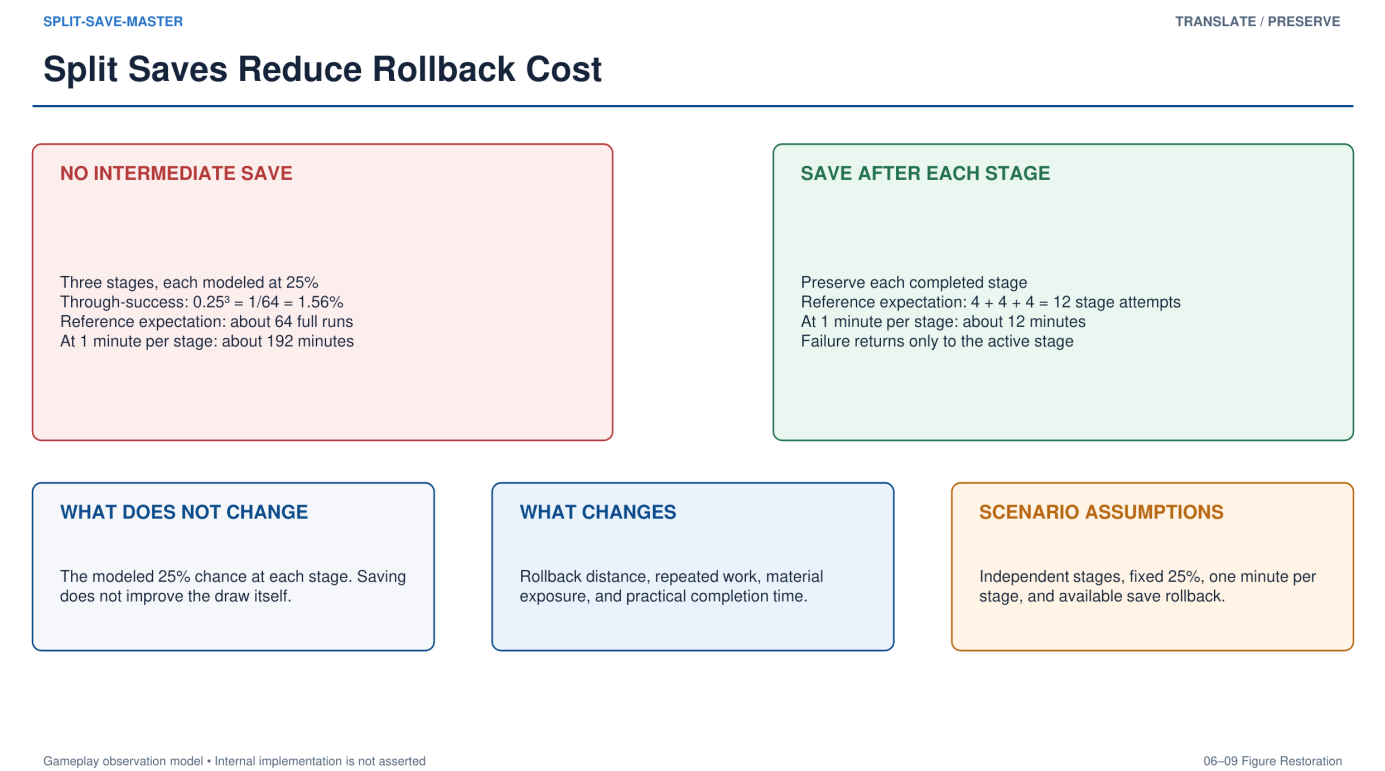


Placement caption

Split-save expected-attempt reduction model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 07-P20-img-008

Master: SPLIT-SAVE-MASTER | Source anchor: Japanese Canonical p.20



Placement caption

Split-save fixed-time comparison
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Candidate count determines the size of the combination space. It does not, by itself, determine the probability distribution over that space.

The RF4SP 50-trial turnip-family observation is direct evidence that equal subset selection cannot be generalized across all observed conditions. The Messhilite 400-trial observation prevents the opposite overgeneralization: one strongly skewed condition does not establish universal weighting.

7-6. Different-type mixing and repeated winning candidates

Adding a different losing candidate increases N without increasing W ; under equal selection, success probability decreases. Adding interchangeable winning copies may increase both N and W .

For example, if three candidates count as winning and one as losing, then $W = C(3,3) = 1$ and $K = C(4,3) = 4$, giving 25% under the equal-selection approximation. If six of seven candidates are interchangeable winning copies, $W = C(6,3) = 20$ and $K = C(7,3) = 35$, giving approximately 57.14%.

Total candidates N	Winning copies A	Winning subsets $C(A,3)$	Approximation $C(A,3)/C(N,3)$
4	3	1	25.00%
5	4	4	40.00%
6	5	10	50.00%
7	6	20	57.14%

This approximation depends on the exact success definition, interchangeability of repeated copies, and equal subset selection. It should not be read as a universal measured success rate.

7-7. Object X and the permutation problem

For Object X inheritance, candidate inclusion may not be sufficient. Because Object X affects materials placed after it, a success condition may also require Object X to occupy the first displayed position.

Two cases must be separated before calculating an order space.

Case 1: three distinct retained candidates

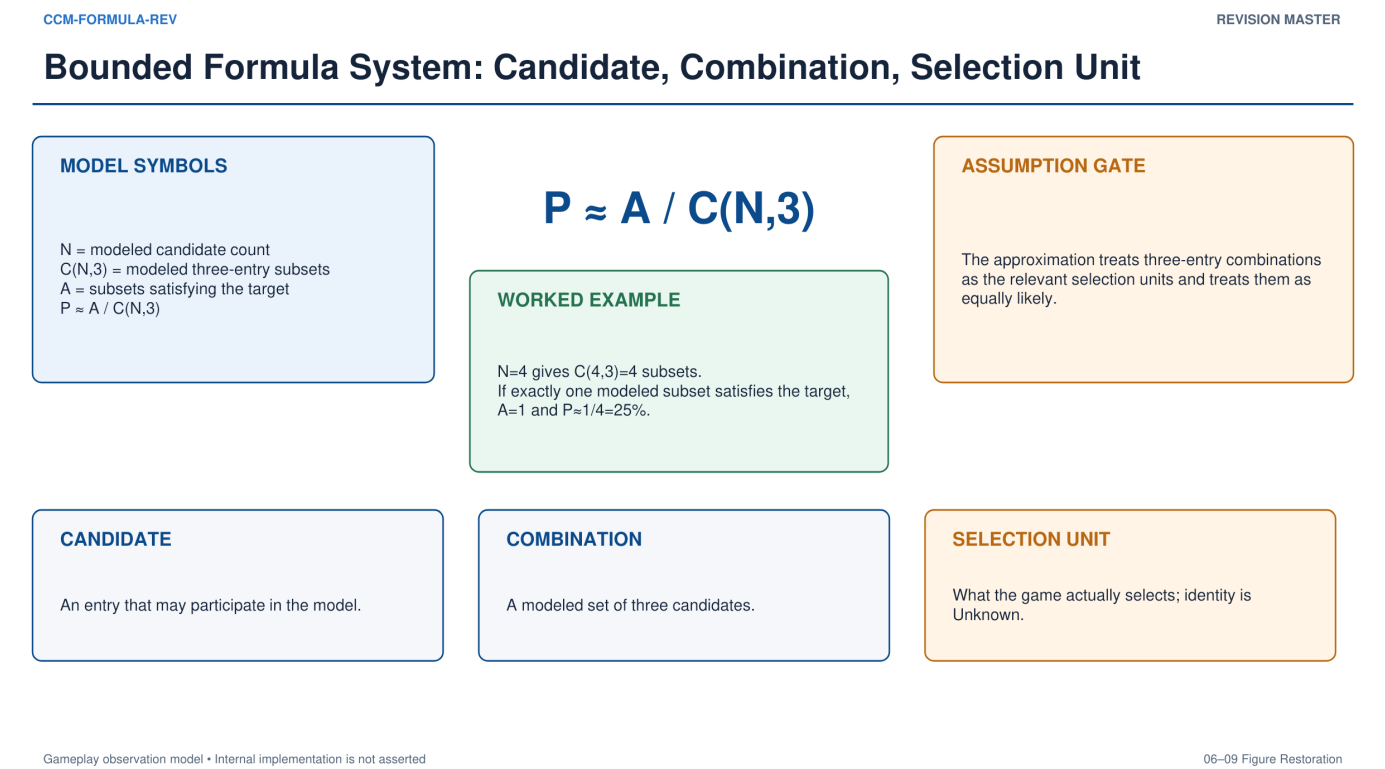
For distinct candidates X , A , and B , there are six distinguishable orders:

XAB , XBA , AXB , ABX , BXA , BAX .

Under an **equal-permutation assumption**:

Figure 07-P23-img-010

Master: CCM-FORMULA-REV | Source anchor: Japanese Canonical p.23

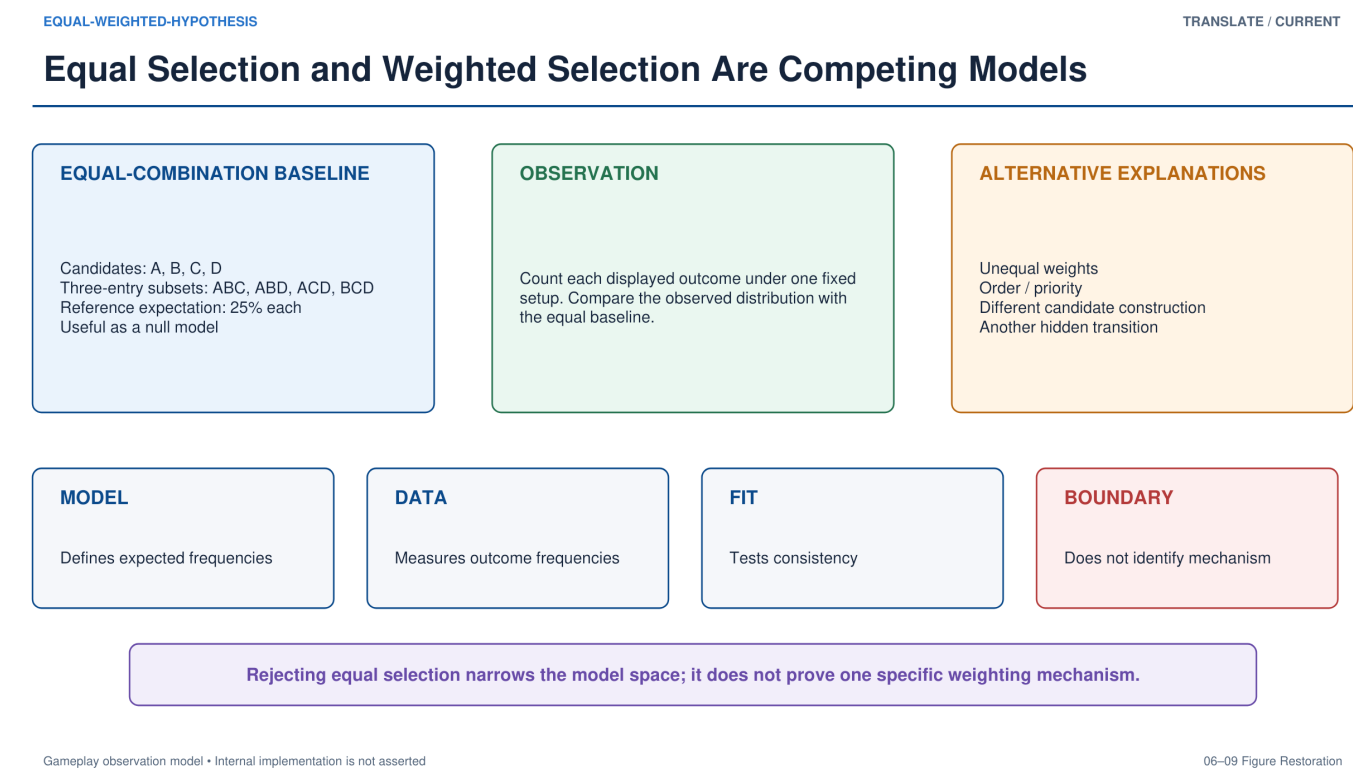


Placement caption

Formula $P=1/C(N,3)$
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 07-P26-img-012

Master: EQUAL-WEIGHTED-HYPOTHESIS | Source anchor: Japanese Canonical p.26



Success definition	Winning orders	Conditional approximation
Exact XAB only	1 of 6	16.67%
XAB or XBA (X first; A/B order irrelevant)	2 of 6	33.33%
Any order of X, A, and B	6 of 6	100%

These values are comparison baselines, not observed RF4SP probabilities.

Case 2: Object X plus two identical materials

The actual 300-trial RF4SP condition used Object X plus two copies of the same material. The two copies are not distinguishable as A and B in the observed output. The directly observable order space is therefore three Object X positions:

XAA, AXA, and AAX.

The test recorded 163 / 92 / 45, or 54.33% / 30.67% / 15.00% (chi-square = 70.58 against an equal one-third position model). The final position was therefore not uniformly distributed under that condition.

This observation does **not** directly test the six-order XAB model. It invalidates using a uniform one-third Object X position assumption for this repeated-material condition. A separate test with two distinguishable materials would be required to estimate all six distinct orders.

Material-condition comparison

The skew was not reproduced across the four other repeated-material conditions examined in RF4SP. Each of those conditions used 50 trials:

Repeated material with Object X	X last / middle / first	chi-square	p
Heavy Spice ×2	17 / 17 / 16	0.04	0.980
Sweet Powder ×2	17 / 17 / 16	0.04	0.980
Sour Drop ×2	20 / 18 / 12	2.08	0.353
Mixed Herbs ×2	16 / 14 / 20	1.12	0.571
Mealy Apple ×2	45 / 92 / 163	70.58	approximately 4.44×10^{-16}

The four 50-trial material conditions are consistent with the equal-position approximation in the limited sense that the null hypothesis was not rejected. This does not prove equal positions. The Mealy Apple condition, tested over 300 trials, showed a strong and differently shaped skew.

Master: MESSHILITE-PRELIMINARY-OBS | Source anchor: Japanese Canonical p.30

MESSHILITE-PRELIMINARY-OBS

Preliminary Observation
The beginning of the Messhite inheritance research

1. Unexpected Observation
A peculiar inheritance behavior was repeatedly observed.

2. Repeated Similar Results
Similar results occurred across multiple attempts and different conditions.

3. Existing Explanation Was Insufficient
Known Mechanisms → Observed Behavior → Mismatch
Standard inheritance mechanics explain most outcomes. The observed results do not fully align with known mechanisms. This gap suggests an additional hidden mechanism.

4. New Hypothesis
The candidate count model may explain the observed behavior.
Candidate Count → Combination Space → Success Rate
This hypothesis may account for the unusual inheritance behavior.

5. Further Investigation
Small Observation → Repeated Testing → Large-scale Validation → Mathematical Interpretation
Initial results were recorded. More tests were conducted under various conditions. Large-scale tests were performed to confirm trends. Results were interpreted using mathematical models.

Key Takeaway
Unexpected observations can reveal hidden game mechanics. Careful observation and repeated validation may transform isolated anomalies into reproducible models.

EXPLANATORY FUNCTION RETAINED

The source figure retains the input materials, crafting record, displayed result, preliminary comparison, and follow-up question.

CURRENT INTERPRETATION

Preserve the observed input/output record. The preliminary difference motivates validation but does not identify the causal transition.

SOURCE DETAIL + CURRENT BOUNDARY

06-09 Figure Restoration

Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 07-P31-img-014

Master: MI-MODEL-OBS-MASTER | Source anchor: Japanese Canonical p.31

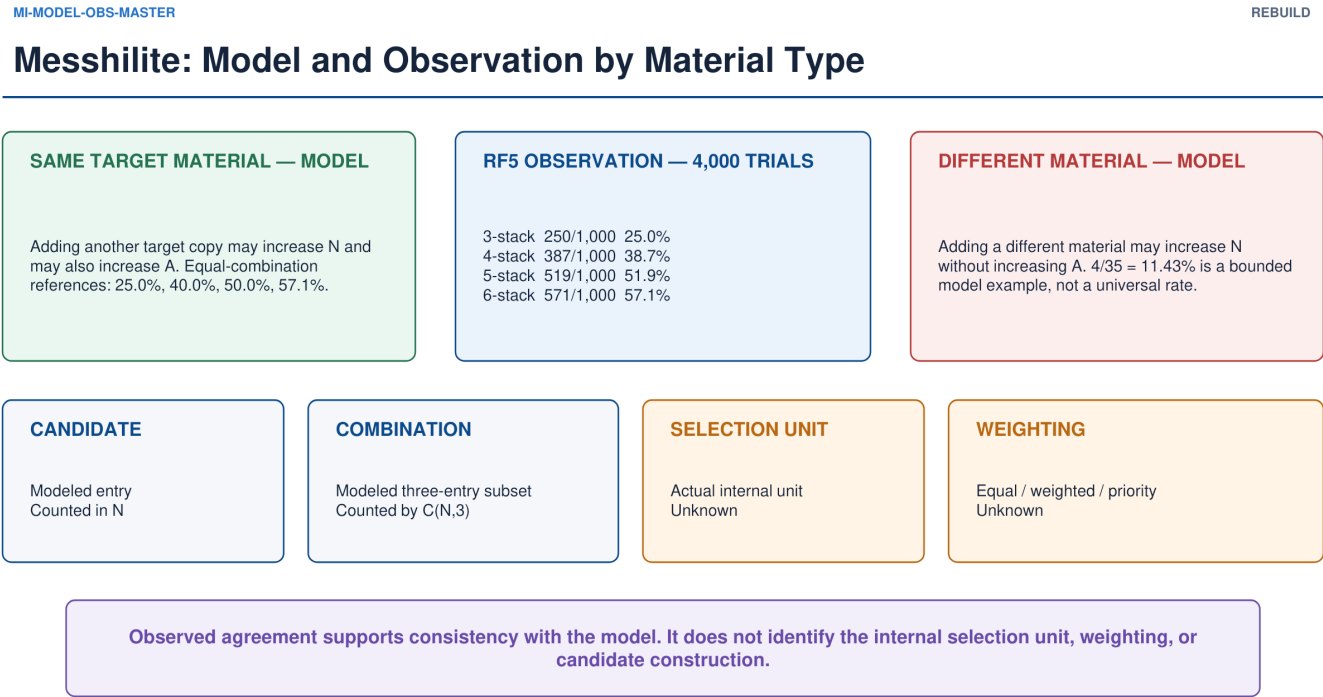
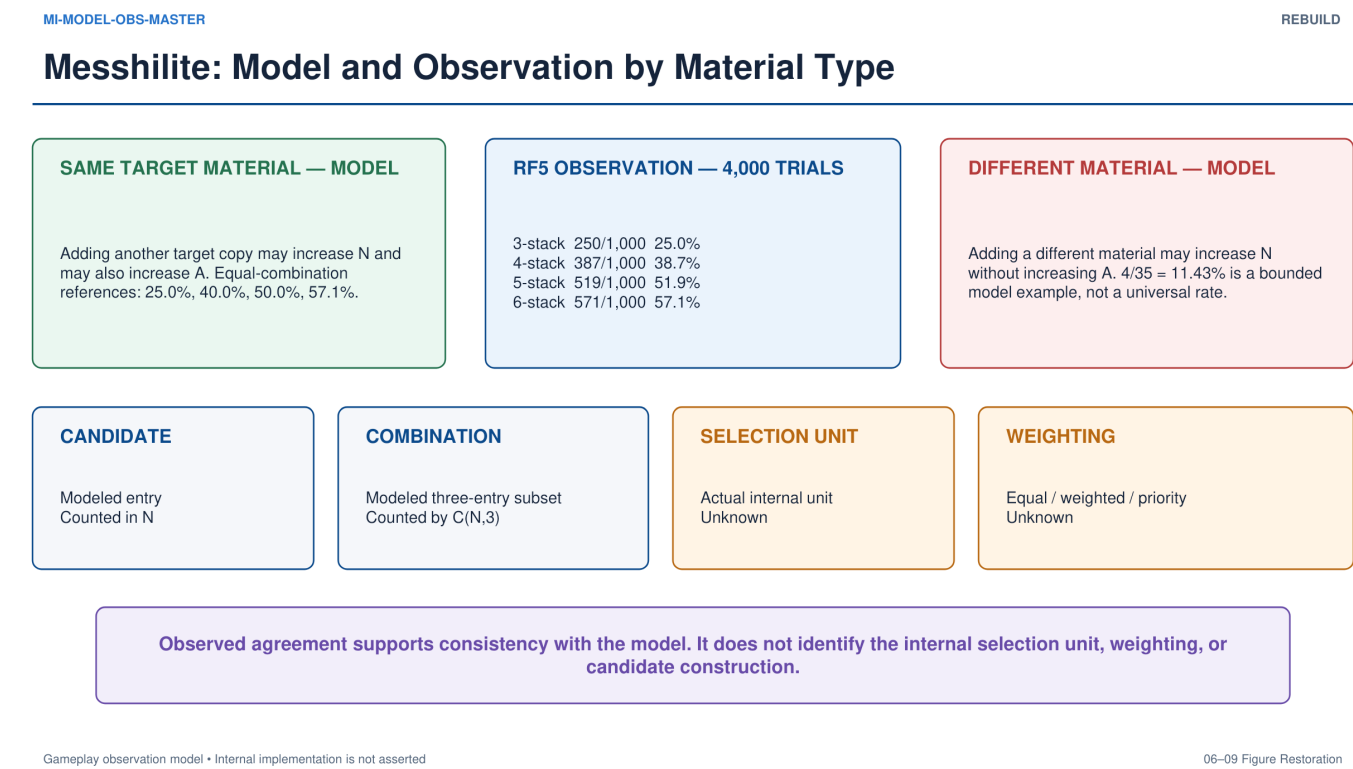


Figure 07-P32-img-015

Master: MI-MODEL-OBS-MASTER | Source anchor: Japanese Canonical p.32



Accordingly, the current evidence supports neither a general claim that RF4SP order is uniformly distributed nor a general claim that it is always biased. The observed bias is **condition-specific**. Whether the difference arises from a material-specific property, candidate registration, an interaction with Object X, or another process remains Unknown.

RF5 showed comparatively stable first-position retention in the tested $N \leq 3$ conditions. RF4SP allowed position variation, but the resulting distribution differed by material condition. This is an observed game-version and condition difference, not proof of a specific order algorithm.

The generalized success decomposition is:

$$P(\text{success}) = P(\text{subset success}) \times P(\text{order success} \mid \text{retained subset})$$

The old simple formula $2/[N(N-1)(N-2)]$ corresponds to a narrow equal-selection/equal-permutation condition where X must be first and two **distinct** target candidates may appear in either order. It does not represent the stricter XAB only condition, it does not directly describe the repeated-material XAA / AXA / AAX experiment, and it is not valid as an empirical RF4SP formula after the observed position skew.

Thus, **candidate formation** and **permutation formation** are separate analytical questions. Whether they are separate internal game procedures remains unresolved.

7-8. Weighted-selection hypothesis and unresolved transitions

Observed deviations from equal selection can be described by multiple competing explanations: candidate-specific weights, registration order, existing-arrangement priority, conditional selection, candidate generation, reselection, or a later ordering rule. Current evidence does not identify which explanation is correct.

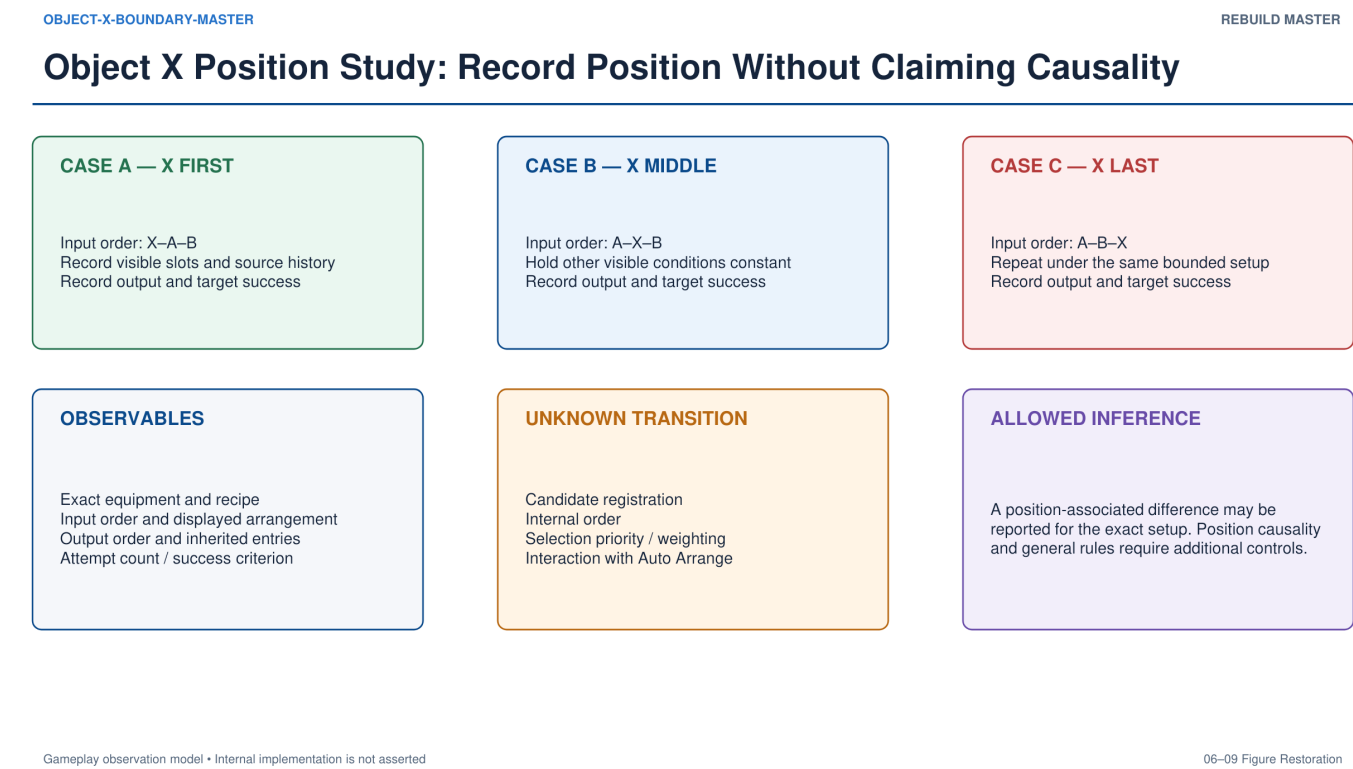
The common observation framework therefore uses the following analytical sequence:

1. Candidate Collection — identify information that may become eligible.
2. Candidate Set — represent the total candidate group.
3. Candidate Combination — enumerate possible retained subsets.
4. Candidate Selection — observe which subset remains.
5. Candidate Permutation — observe the final order when order matters.
6. Observed Outcome — record what the game actually produced.

Set theory organizes candidate membership. Combinatorics organizes possible subsets and permutations. Probability describes how frequently outcomes occur. None of these layers, by itself, claims to reproduce the game's internal implementation.

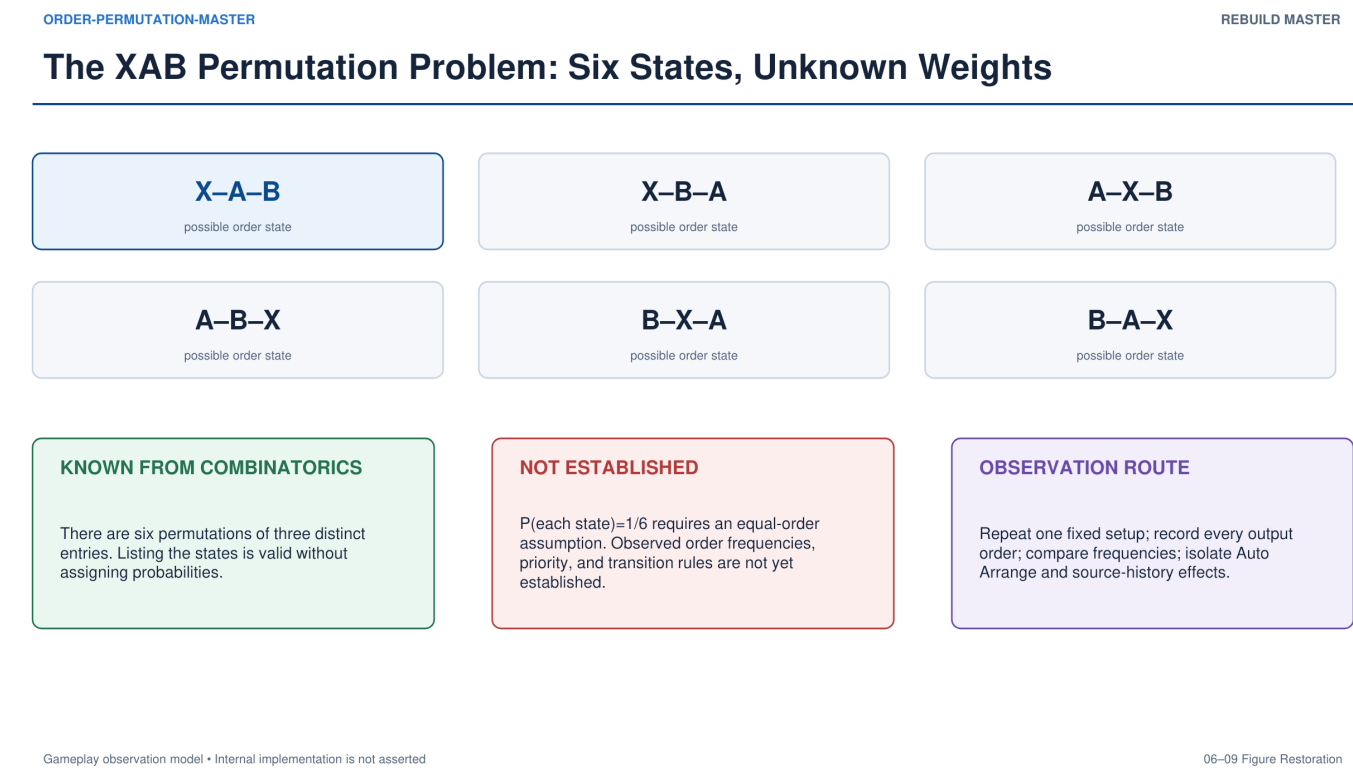
Unresolved questions

- Why did the RF4SP turnip-family condition show strong subset skew while the Messhilite 400-trial condition did not?



Placement caption

Object X position comparison
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

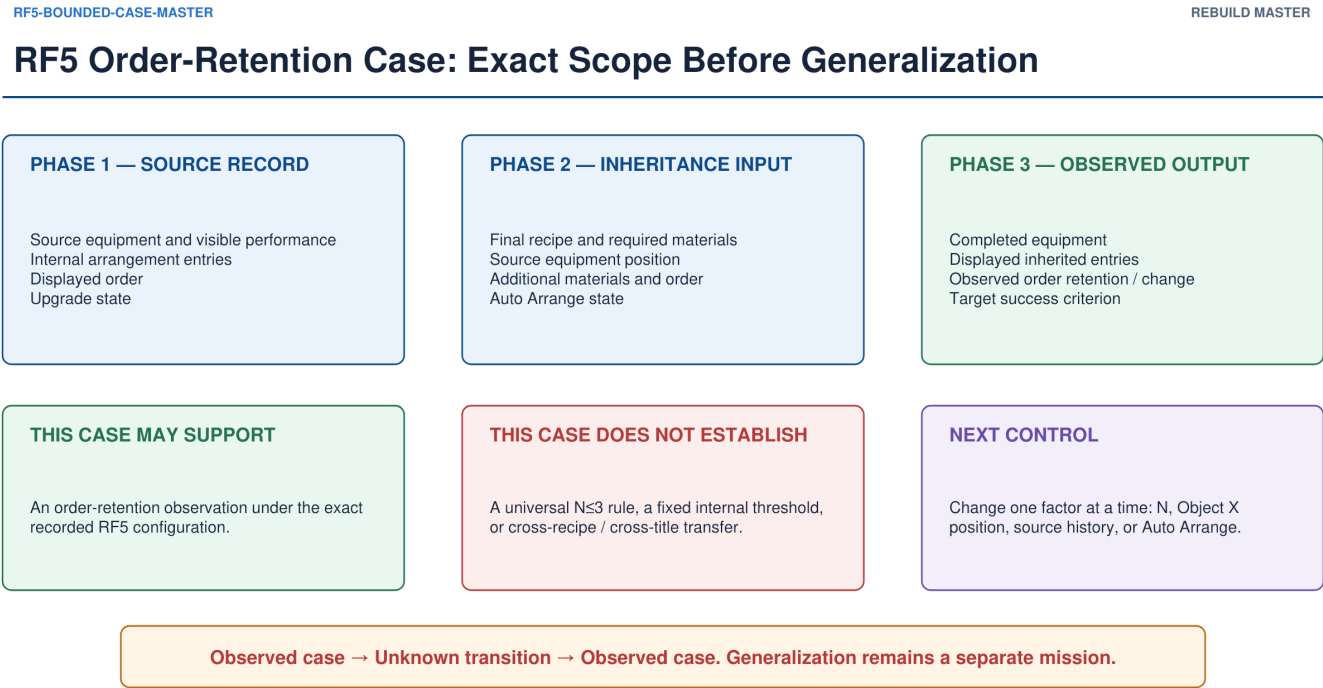


Placement caption

Permutation problem for XAB
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

Figure 07-P42-img-020

Master: RF5-BOUNDED-CASE-MASTER | Source anchor: Japanese Canonical p.42



Placement caption

RF5 $N \leq 3$ order-retention model
Evidence boundary: the figure separates Observation, Model, and Unknown Transition as applicable.

- Why did the RF4SP Mealy Apple condition show strong Object X position skew while the four other repeated-material conditions remained consistent with the equal-position approximation?
- Which factor produced the RF4SP Object X position skew?
- Does candidate registration order affect subset selection, final order, or both?
- When does RF5 retain input order, and when does that behavior break down?
- How do occupied internal arrangement slots and Auto Arrange alter the effective candidate set?
- Can subset selection and final ordering be experimentally isolated without assuming an internal processing sequence?

Chapter boundary

This chapter is an observation-based mathematical framework. It refines the Candidate Count Model from a universal-looking success-rate formula into:

- a model of combination-space growth;
- a conditional equal-selection approximation;
- a separate permutation condition when order matters;
- an explicit Unknown boundary around the actual distribution and internal transitions.

The Candidate Count Model is **not falsified**. Its applicability has been refined. The selection mechanism and order mechanism remain **Unknown**.

Master: CUMULATIVE-PROBABILITY-CURRENT | Source anchor: Japanese Canonical p.50

REUSE / TRANSLATE

Success Probability Curve

As the candidate pool expands, successful combinations become increasingly diluted, reducing the probability of obtaining a specific inheritance result.

The graph illustrates the relationship between the number of candidates (N) and the success probability. The y-axis represents Success Probability on a logarithmic scale from 0.1% to 100%. The x-axis represents the Candidate Count (N) from 3 to 8. The curve shows that as N increases, the success probability drops significantly. The graph is divided into three regions: Stable Region (N=3 to 4), Candidate Explosion (N=4 to 6), and Combination Dilution (N=6 to 8).

Candidate Count (N)	Combinations	Success Probability (C(N,3))
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More Candidates
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More Possible Combinations
The number of possible combinations increases exponentially.

Lower Chance
The chance of obtaining a specific combination drops rapidly.

Key Takeaway

Increasing the candidate count expands the combination space much faster than most players intuitively expect. As a result, even when the desired inheritance remains possible, its relative probability rapidly decreases because it occupies a smaller fraction of all possible combinations.

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